

MRPL Refinery — Pressure Safety Valve (PSV) Testing and Inspection Record

Document Control

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Scope

This record covers pop-testing of pressure safety valves on pressure vessels and columns in the Fluid Catalytic Cracking Unit (FCCU) and Crude Distillation Unit (CDU), performed during the Q3 2026 turnaround per statutory OISD-116 requirement of testing at intervals not exceeding 24 months, or per unit-specific risk-based inspection (RBI) schedule.

Test Results Summary

PSV Tag	Protected Equipment	Set Pressure (bar g)	As-Found	As-Left	Result
PSV-2201	V-220 Main Fractionator	4.2	4.15	4.20	Pass
PSV-2214	C-101 Discharge	40.0	37.8	40.0	Fail - Reset
PSV-3305	E-305 Shell Side	12.5	12.4	12.5	Pass
PSV-1102	V-110 Reactor	18.0	18.1	18.0	Pass

Finding — PSV-2214

PSV-2214, protecting the discharge line of Compressor C-101, was found to lift at 37.8 bar g against a set pressure of 40.0 bar g (5.5 percent below set point). This is within the OISD-116 acceptable as-found tolerance of minus 3 to plus 3 percent only marginally exceeded; the valve was bench-reset to 40.0 bar g and re-tested twice to confirm repeatability. Root cause was attributed to spring relaxation after approximately 22 months in service. No process safety event occurred as the valve did not lift in service during this period.

Recommended Action

- PSV-2214 has been reset and returned to service; log updated in PSV master register.
- Reduce re-test interval for this valve to 18 months given early spring relaxation trend.
- Review spring material specification with OEM for future replacement units on high-cycle service.